The Z-Plane Covariance Operator of the Hilbert Curve: A Tridiagonal Toeplitz Spectrum That Correlates With, But Cannot Equal, the Riemann Zeros

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Abstract

We study a self-adjoint operator $C_D^{(n)}$ obtained by projecting the prime indicator function onto the Z-axis of a D-dimensional Hilbert space-filling curve of order n and computing the resulting spatial covariance between axis-aligned slabs ("Z-planes"). We prove that when $C_D^{(n)}$ is built from face-adjacency of the ambient grid, it is *exactly* tridiagonal Toeplitz, independent of D, n, and the choice of indicator function, giving the closed-form spectrum $\lambda_k = c_0 + 2c_1 \cos(\pi k/(n+1))$. Empirically, for the prime indicator, the largest eigenvalues of $C_D^{(n)}$ correlate with the imaginary parts of the first 64 non-trivial zeta zeros at $|r|$ up to 0.990 ($D = 4$, spatial construction, order 4), a correlation we independently re-derive from the underlying covariance data with a properly randomized permutation test (previous reported significance figures rested on a non-randomized shuffle). We show the correlation is a finite-range artifact: the Taylor expansion of the cosine spectrum reproduces the empirically observed quadratic approximation to $|\lambda(\gamma)|$, but the operator's spectrum is bounded in $[c_0 - 2|c_1|, c_0 + 2|c_1|]$ for every n, while the zeta zero counting function grows as $N(T) \sim (T/2\pi)\log(T/2\pi e)$. No finite- or infinite-order member of this operator family can therefore satisfy the Hilbert–Pólya correspondence exactly, regardless of D or n. We give a precise statement of what spectral properties such an operator would need to have, and report the reproducibility issues we found and corrected in re-deriving these results.

1 Introduction

The Hilbert–Pólya conjecture proposes that the imaginary parts γ_k of the non-trivial zeros of $\zeta(s)$ are the eigenvalues of a self-adjoint operator on some Hilbert space; if such an operator can be exhibited, the Riemann Hypothesis (RH) follows from self-adjointness alone. No computationally explicit candidate operator has been broadly accepted.

This paper studies one geometrically motivated candidate: project the indicator function of the primes onto axis-aligned slabs of a D-dimensional Hilbert curve, and compute the empirical covariance between slabs induced by grid adjacency. This construction has two independent appeals. First, the Hilbert curve is measure-preserving and its slab decomposition is canonical, so the resulting operator has no free parameters beyond D and the curve order n. Second, as we show in Section 3, the operator's structure is derivable in closed form: it is not merely diagonalizable numerically but tridiagonal Toeplitz *by construction*, with a spectrum given by an elementary trigonometric formula independent of D and of any arithmetic property of the primes.

Our contribution is threefold:

1. We prove the tridiagonal Toeplitz structure and derive its spectrum in closed form (Theorem 1), explaining an empirical observation from earlier numerical work [10] as a consequence of face-adjacency alone, independent of dimension.

2. We give an independent, from-scratch numerical re-derivation of the eigenvalue–zeta-zero correlation reported for this construction, using a properly randomized permutation test. Two irregularities in the original analysis pipeline are identified and corrected in the process (Section 7): a non-randomized "permutation" test, and a curve-implementation/documentation mismatch for the 3D case. The correlation itself survives both corrections essentially unchanged.

3. We give a precise argument, in the language of spectral density, for why this operator family cannot satisfy the Hilbert–Pólya correspondence at any order or dimension, and state the density condition any successor construction would need to satisfy.

2 Construction

2.1 The Hilbert curve and its Z-planes

Let $H_D : \{0, \ldots, 2^{Dn} - 1\} \to \{0, \ldots, 2^n - 1\}^D$ be the D-dimensional Hilbert curve of order n, realized via the recursive Butz construction [1, 2]: at each of n recursion levels, D bits of the index select one of 2^D sub-hypercubes, with a Gray-code map and an orientation state $(e, d) \in \{0, \ldots, 2^D - 1\} \times \{0, \ldots, D - 1\}$ propagated across levels to keep consecutive sub-hypercubes face-adjacent. We verified this implementation directly: the fraction of consecutive integers $k, k+1$ whose images differ by a single face-step (ℓ_1 distance 1) is 89.3% ($D = 3$), 94.2% ($D = 4$), 97.0% ($D = 5$), and 98.5% ($D = 6$) at order 4, consistent with the boundary-crossing loss $1 - (1 - 2^{1-n})^D$ predicted by the construction.

The Z-plane of a cell $(x_1, \ldots, x_D)$ is $z = x_3$. This partitions the 2^{Dn} cells into 2^n slabs of $2^{(D-1)n}$ cells each, independent of D: increasing the ambient dimension only enlarges the slabs, not their number.

2.2 Two covariance constructions

Let $f : \{0, \ldots, 2^{Dn} - 1\} \to \{0, 1\}$ be the prime indicator ($f(k) = 1$ iff k is prime), and let μ_z be the mean of f over cells in plane z. We consider two ways of coupling planes.

Curve-adjacent. For each consecutive pair $(k, k+1)$ along the curve, with Z-planes z_1, z_2:

$$C^{\text{curve}}[z_1, z_2] = \frac{1}{N_{z_1 z_2}} \sum_{(k,k+1)} (f(k) - \mu_{z_1})(f(k+1) - \mu_{z_2}). \tag{1}$$

Spatial. For each cell c and each of its $2D$ face-neighbors c' in the ambient grid, with Z-planes z_1, z_2 of c, c':

$$C^{\text{spatial}}[z_1, z_2] = \frac{1}{N_{z_1 z_2}} \sum_{c} \sum_{c'} (f(d_c) - \mu_{z_1})(f(d_{c'}) - \mu_{z_2}), \tag{2}$$

where d_c is the Hilbert index of cell c. The spatial construction measures genuine geometric clustering; the curve-adjacent construction measures clustering along the traversal order, which coincides with spatial adjacency only 89–98.5% of the time (Section 2.1).

Both matrices are symmetrized ($C \leftarrow (C + C^\top)/2$) and are real, so their eigenvalues are real. We write $\lambda_1, \ldots, \lambda_n$ for the eigenvalues sorted by $|\lambda_i|$ descending.

3 The Tridiagonal Toeplitz Theorem

Theorem 1. *For any $D \geq 3$, any order n, and any function f on the cells of the D-dimensional Hilbert curve, the spatial covariance matrix C^{spatial} satisfies $C^{\text{spatial}}[z_1, z_2] = 0$ for all $|z_1 - z_2| > 1$.*

Proof. A face-neighbor of a cell changes exactly one coordinate by ± 1. Of the $2D$ face-neighbors of a cell $(x_1, \ldots, x_D)$, exactly $2(D-1)$ change a coordinate other than x_3 and leave the Z-plane $z = x_3$ unchanged; the remaining 2 change x_3 by ± 1, moving to an adjacent plane. No face-neighbor changes x_3 by 2 or more. Hence every term contributing to $C^{\text{spatial}}[z_1, z_2]$ has $|z_1 - z_2| \leq 1$, and the sum defining entries with $|z_1 - z_2| > 1$ is empty. □

This is a statement about the adjacency graph of the grid, not about f or about primes; it holds verbatim for a random f and is therefore also a practical diagnostic for whether a candidate space-filling curve implementation is genuinely face-adjacent (Section 7 gives an example where it was not).

Proposition 1. C^{spatial} *is (interior-)Toeplitz:* $C[z, z] \approx c_0$ *and* $C[z, z \pm 1] \approx c_1$ *for all* z*, with* $|c_1/c_0| \to 1$ *as* $n \to \infty$*.*

This follows because the diagonal averages the same-plane contribution over $2(D-1)$ neighbor pairs and the off-diagonal averages the cross-plane contribution over 2 neighbor pairs; both estimate $\text{Var}[f]$ after normalization by pair count, so their ratio tends to 1 independent of D. Table 1 confirms this for the prime indicator in $D = 4$.

n	c_0	c_1	$\lvert c_1/c_0\rvert$	$\max \lvert C[z_1, z_2]\rvert_{\lvert z_1 - z_2\rvert > 1}$
4	-9.940×10^{-3}	-9.997×10^{-3}	1.0057	0
5	-6.318×10^{-3}	-6.305×10^{-3}	0.9980	0
6	-4.195×10^{-3}	-4.235×10^{-3}	1.0097	0
7	-3.048×10^{-3}	-3.048×10^{-3}	0.99995	0

Table 1: 4D spatial covariance for the prime indicator, orders 4–7 (re-measured directly from the covariance matrices). $c_2, c_3, \ldots$ are zero to floating-point precision at every order; the corresponding curve-adjacent matrices at the same orders have $\max |C[z_1, z_2]|_{|z_1 - z_2| > 1} \in [0.004, 0.03]$, confirming the tridiagonal property is specific to the spatial construction.

Corollary 1 (Grenander–Szegő [3])**.** *The eigenvalues and eigenvectors of* C^{spatial} *are*

$$\lambda_k = c_0 + 2c_1 \cos\left(\frac{\pi k}{n+1}\right), \qquad v_k[j] = \sin\left(\frac{\pi k j}{n+1}\right), \qquad k, j = 1, \ldots, n. \tag{3}$$

We verified Eq. (3) directly against numerical diagonalization of the four matrices in Table 1, with eigenvalues ordered to match the monotone-in-k form of Eq. (3) (i.e., ascending, *not* sorted by $|\lambda|$ – the two orderings coincide only for $k \lesssim 2n/3$; see Remark 1). Table 2 reports the fit.

n (order)	matrix size	Pearson r	MAD (raw)
4	16	0.999859	1.71×10^{-4}
5	32	0.999920	2.65×10^{-4}
6	64	0.999961	1.42×10^{-4}
7	128	0.999976	1.11×10^{-4}

Table 2: Closed-form spectrum (Eq. 3) vs. numerically diagonalized eigenvalues, 4D spatial covariance. Agreement improves with n, consistent with the boundary-cell fraction ($\sim 6\%$ at $n = 7$) being the dominant source of residual error.

Remark 1. *Eq.* (3) *is monotone increasing in k (since $c_0 = c_1 < 0$), running from $3c_0$ at $k = 1$ to $-c_0$ at $k = n$. Its* magnitude $|\lambda_k|$ *is therefore not monotone: it decreases from $k = 1$ to a zero crossing near $k \approx 2n/3$ and then increases again with the opposite sign. The two natural orderings – by k, or by $|\lambda|$ descending – agree only on the first branch. This distinction matters: fitting Eq.* (3) *requires the natural k-ordering (Table 2); comparing against the zeta zeros (Section 4) uses the $|\lambda|$-descending ordering, following the original construction, restricted to the first branch ($k \leq 64 \ll 2n/3$ for $n \geq 96$). Conflating the two orderings when n is not $\gg 64$ silently produces a large, spurious drop in the fitted correlation; we encountered this exact error while independently re-deriving Table 2 and record it here as a caution for future work with this construction.*

4 Correlation with the Zeta Zeros

4.1 Method

For each covariance matrix we compute eigenvalues, sort by $|\lambda_i|$ descending, take the first $k = \min(64, n)$, and compare against the first k zeta zeros $\gamma_1 < \cdots < \gamma_k$ (LMFDB [9]) via: Pearson r between $|\lambda_i|$ and γ_i; MAD between both sequences independently min–max normalized to $[0, 1]$; and an empirical p-value from 2×10^4 trials of a permutation test that independently shuffles the eigenvalue vector using a seeded pseudo-random generator, re-drawn on every trial, and reports the fraction of shuffles with $|r_{\text{perm}}| \geq |r_{\text{obs}}|$.

4.2 Results

Table 3 gives results across $D \in \{3, 4, 5, 6\}$, using whichever construction (curve-adjacent or spatial) is available for each D in the underlying data. All values were computed independently from the covariance matrices, not taken from prior reports.

D	Construction	Order n	Pearson r	MAD	perm. p	off-tridiag.
3	curve-adj.	6	-0.927	0.546	$< 5 \times 10^{-5}$	4.9×10^{-2}
3	curve-adj.	9	-0.890	0.512	$< 5 \times 10^{-5}$	3.4×10^{-2}
3	curve-adj.	11	-0.919	0.504	$< 5 \times 10^{-5}$	1.2×10^{-2}
4	curve-adj.	6	-0.866	0.518	$< 5 \times 10^{-5}$	3.0×10^{-2}
4	spatial	6	-0.988	0.538	$< 5 \times 10^{-5}$	0
4	spatial	7	-0.961	0.500	$< 5 \times 10^{-5}$	0
5	curve-adj.	4	-0.981	0.557	$< 5 \times 10^{-5}$	6.8×10^{-3}
5	curve-adj.	5	-0.943	0.520	$< 5 \times 10^{-5}$	1.4×10^{-2}
6	curve-adj.	4	-0.971	0.544	$< 5 \times 10^{-5}$	2.6×10^{-3}

Table 3: Eigenvalue–zeta-zero correlation, independently reconstructed. "perm. p" is from 2×10^4 properly randomized shuffles (0 of 2×10^4 met or exceeded $|r_{\text{obs}}|$ in every row shown). Full results across 32 matrices (orders 2–11, all available D and both constructions) are in the supplementary script output.

The strongest and most consistent correlation is $D = 4$ spatial ($r = -0.990$ to -0.961 across orders 4–7, exactly tridiagonal at every order). The 3D and 4D curve-adjacent constructions perform comparably to each other at matched order ($r \approx -0.87$ to -0.93); dimension alone does not predict correlation strength once the construction is fixed (Table 4).

Remark 2. *The Spearman rank correlation is -1 in essentially every case in Table 3. This is not independent evidence: both $|\lambda_i|$ (sorted) and γ_i (sorted) are monotone sequences of opposite orientation, which forces $\rho = -1$ regardless of any deeper relationship. Only the Pearson r on*

Construction (order 6)	D	r	Off-tridiagonal
Curve-adjacent (original impl.)	3	-0.927	4.9×10^{-2}
Curve-adjacent (Butz, 89% adj.)	3	-0.943	2.6×10^{-2}
Curve-adjacent (Butz, 94% adj.)	4	-0.866	3.0×10^{-2}
Spatial (Butz, 94% adj.)	4	-0.988	0

Table 4: Order-6 comparison across constructions. Two independent 3D implementations (differing in curve-adjacency percentage, 50% vs. 89%) agree to within 0.016 in r, confirming the correlation does not depend sensitively on curve-implementation details. 4D only outperforms 3D once the spatial construction is used; the curve-adjacent constructions in 3D and 4D are comparable.

the unsorted-within-rank *values, and the permutation test, carry information about whether the specific magnitudes align beyond monotonicity.*

4.3 Origin of the quadratic approximation

Combining Eq. (3) with the Riemann–von Mangoldt counting function $N(T) = (T/2\pi)\log(T/2\pi e) + 7/8 + O(\log T)$ and identifying the eigenvalue rank k with $N(\gamma_k)$, the small-angle expansion $\cos(\pi k/(n+1)) \approx 1 - \pi^2 k^2/2(n+1)^2$ gives

$$|\lambda_k| \approx |c_0 + 2c_1| - \frac{c_1 \pi^2}{(n+1)^2} N(\gamma_k)^2, \tag{4}$$

which is quadratic in $N(\gamma)$ and, since $\log^2 \gamma$ varies by only a factor of ~ 3.5 over $\gamma \in [14, 170]$ (the first 64 zeros), approximately quadratic in γ itself over this finite range. This reproduces the previously reported empirical fit $|\lambda(\gamma)| \approx a\gamma^2 + b\gamma + c$ (residual MAD ≈ 0.007) as a second-order Taylor artifact of the cosine spectrum, not as an independent regularity.

5 Why the Correspondence Cannot Be Exact

Proposition 2 (Spectral class mismatch)**.** *For every $D \geq 3$ and every order n, the spectrum $\{\lambda_k\}_{k=1}^n$ of $C_D^{(n),\text{spatial}}$ satisfies $\lambda_k \in [c_0 - 2|c_1|, c_0 + 2|c_1|]$, an interval of width $4|c_1| \to 0$ as the sample size $2^{(D-1)n}$ per plane grows (since $c_0, c_1 = O(1/\sqrt{\text{plane size}})$ by the CLT for a Bernoulli indicator). The zeta zero sequence satisfies $\gamma_k \to \infty$ with counting function $N(T) \sim (T/2\pi)\log(T/2\pi e)$. No affine or polynomial recalibration ϕ with $\gamma_k = \phi(\lambda_k)$ for all k and all n can exist, because the left side is unbounded in k for fixed ϕ while the right side remains confined to a shrinking bounded interval.*

This is stronger than the observation that correlation is not identity: it identifies the obstruction as a property of *spectral density*, not of fit quality. A bounded, order-n operator with $O(2^n)$ eigenvalues packed into an interval of width $O(2^{-(D-1)n/2})$ has eigenvalue spacing shrinking exponentially in n at fixed position in the spectrum, whereas the zeta zero spacing $\gamma_{k+1} - \gamma_k \sim 2\pi/\log\gamma_k$ shrinks only logarithmically. Matching the two counting functions asymptotically – a necessary condition for any Hilbert–Pólya candidate – would require an operator whose spectrum literally diverges as $n \to \infty$, which $C_D^{(n)}$, being a normalized covariance (entries bounded by $\mathrm{Var}[f] \leq 1/4$), cannot do by construction.

Remark 3. *This rules out this specific operator family, constructed from any D, any order, and either adjacency rule; it does not address whether some other function f of the Hilbert curve,* *or an unnormalized construction with entries growing in n, could satisfy the density condition. We regard Proposition 2 as delimiting the search space rather than closing it.*

6 Related Approaches

The Z-plane covariance operator is not the first construction to correlate strongly with the zeta zeros over a finite or asymptotic regime while falling short of an exact correspondence. Several established constructions in the mathematical-physics and noncommutative-geometry literature exhibit the same qualitative pattern: an identifiable structural property, not merely insufficient fit quality, is what prevents closure into a proof. We compare four of them to the obstruction identified in Proposition 2.

Fractal potential fits (Wu–Sprung). Wu and Sprung [5] constructed a one-dimensional Schrödinger potential via inverse scattering whose low-lying eigenvalues reproduce the first several hundred Riemann zeros, at the cost of the potential itself being fractal (box-counting dimension 3/2). They note explicitly that *any* finite number of low-lying zeros can be reproduced by adding fluctuations to the smooth background potential – i.e., the construction is an inverse-scattering fit to a prescribed finite target, not a forward derivation from an independent principle. This is the closest structural analogue to the present work: both constructions achieve strong finite-range agreement precisely because they have enough free parameters (fluctuation terms; choice of D, n, and adjacency rule) relative to the 64–500 zeros being matched, and neither has been shown to constrain the zeros outside the fitted range.

The $H = xp$ model (Berry–Keating). Berry and Keating [6] proposed the classical Hamiltonian $H = xp$, regularized by restricting phase space to $|x| > \ell_x$, $|p| > \ell_p$ with $\ell_x \ell_p = 2\pi\hbar$. The resulting semiclassical counting function $N_{BK}(E) = (E/2\pi)(\log(E/2\pi) - 1) + 1$ reproduces the leading $T \log T$ growth of the true zeta zero counting function *by construction* – unlike $C_D^{(n)}$, whose bounded spectrum cannot do so at any order (Proposition 2). What the xp model lacks is rigor at the operator level: xp is not essentially self-adjoint on $L^2(\mathbb{R})$, and the phase-space cutoff has no first-principles justification beyond reproducing the correct asymptotic density. The two constructions thus fail in complementary ways – correct density without a rigorous operator, versus a rigorous (finite-matrix) operator with the wrong density.

Landau levels (Sierra–Townsend). Sierra and Townsend [7] relate the zeros to the spectrum of a charged particle in a magnetic field on the hyperbolic half-plane, giving another physically motivated semiclassical route to the counting function with similar caveats about rigor and completeness.

Noncommutative spectral realization (Connes). Connes [8] gives a spectral interpretation of the critical zeros as an absorption spectrum in a trace formula over the adèle class space, under which RH becomes equivalent to a positivity statement about a trace pairing. This sidesteps the density obstruction entirely – the construction is exact, not a finite-range fit – but reduces RH to an equally open positivity conjecture rather than resolving it.

The recurrence of this pattern across independently motivated constructions suggests the density obstruction (or a close relative of it – rigor, completeness) is a generic hazard for finite or semiclassical Hilbert–Pólya candidates, rather than a defect specific to the Hilbert-curve construction studied here.

7 Reproducibility

The results above were re-derived independently from the covariance matrices, rather than taken from the pipeline that originally produced them, and two issues were found and corrected in the process.

Permutation test. The original significance test for the eigenvalue–zeta-zero correlation used a shuffle map that was a fixed function of array index rather than a call to a random-

Construction	Correct density growth	Rigorous operator	Status
$C_D^{(n)}$ (this paper)	No (bounded)	Yes (finite matrix)	Ruled out, all D, n
Wu–Sprung fractal potential	Only by fitting	Yes	Fit, not derived
Berry–Keating $H = xp$	Yes (leading term)	No (needs regularization)	Conjectural
Sierra–Townsend Landau levels	Partial	Semiclassical	Conjectural
Connes spectral realization	Yes (exact)	Yes (adèle space)	Reduces to open positivity

Table 5: The obstruction identified in this paper (bounded vs. unbounded spectral density) is one of several distinct, well-documented reasons a zeta-zero-correlating construction can fail to yield a proof. No entry in this table has all three of: correct asymptotic density, rigorous operator status, and an unconditional resolution of RH.

number generator, and did not reset the array between trials – so repeated "trials" were repeated compositions of one fixed permutation, not independent draws. We replaced this with a seeded Mersenne-Twister shuffle re-drawn every trial (Section 4). The qualitative conclusion is unchanged ($p < 5 \times 10^{-5}$ throughout Table 3), but the quantitative significance figures reported before this correction were not meaningful.

Curve implementation vs. documentation. The 3D results in Table 4 come from two different curve implementations: an original ad-hoc construction achieving 50% face-adjacency, and a Butz-algorithm implementation achieving 89.3%. Prior documentation described the 3D results as using the latter; the pipeline that generated some of the reported 3D numbers in fact used the former. We verified directly (by tracing consecutive-index face-adjacency) that both implementations are in use in different parts of the codebase, and confirm in Table 4 that the discrepancy does not materially affect the reported correlation ($\Delta r = 0.016$ at order 6) – but it should be corrected as a matter of documentation accuracy independent of its numerical impact.

Segmented sieve bug ($D = 5, 6$ and the standalone $D = 3, 4$ cross-check). The prime sieve used to generate the $D \in \{3, 4, 5, 6\}$ covariance matrices independent of the main pipeline contained a classical segmented-sieve error: the inner loop marked composites starting from the smallest multiple of each small prime p that was $\geq$ low, without excluding the trivial multiple p itself. In the first segment (low = 3), this multiple equals p for every small prime, so every prime up to $\sqrt{\text{limit}}$ was marked composite and silently dropped – 96 missing primes at order 6 ($D = 3$), confirmed by comparison against a reference sieve. We fixed this by clamping the starting multiple to p^2 (standard practice, apparently dropped during a refactor), verified the fix against a reference implementation across seven limits spanning single- and multi-segment cases, and regenerated all sixteen affected covariance matrices. The corrected data changes r by < 0.005 at order ≥ 4 for every D (Table 6) – small enough that none of the values reported in Table 3 or Table 4 change at the precision shown – but by up to 0.074 at order 2–3, where a few hundred missing small primes are a large fraction of a total prime count in the low thousands. This is the expected signature of a bug that drops a roughly fixed number of small primes: its relative effect shrinks as the prime count grows with order.

Theorem 1 was used as an internal consistency check throughout: a genuinely face-adjacent curve implementation forces $\max_{|z_1 - z_2| > 1} |C[z_1, z_2]| = 0$ to floating-point precision under the spatial construction, while a buggy implementation with silently reduced adjacency produces detectably nonzero far off-diagonal entries even when its overall bijection property (each index visited once) is intact. This diagnostic is cheaper than auditing consecutive-pair adjacency directly and is, in our view, the most practically useful byproduct of Theorem 1. It did not catch the sieve bug above, since that bug affects which cells are prime, not the curve's adjacency structure – the two diagnostics are complementary, not overlapping.

File	Order	r (buggy sieve)	r (fixed)	Δr
cov_3d_n2.json	2	−0.9963	−0.9225	+0.074
cov_3d_n3.json	3	−0.9654	−0.9865	−0.021
cov_5d_n3.json	3	−0.9692	−0.9539	+0.015
cov_4d_n2.json	2	−0.9464	−0.9624	−0.016
cov_3d_n6.json	6	−0.9432	−0.9434	−0.0003
cov_4d_n6.json	6	−0.8660	−0.8663	−0.0003
cov_5d_n5.json	5	−0.9429	−0.9430	−0.0001
cov_6d_n4.json	4	−0.9714	−0.9714	−0.0000

Table 6: Effect of the sieve fix on Pearson r, low orders (top, large relative effect) vs. the orders actually used in Tables 3–4 (bottom, negligible effect). Full before/after comparison across all 16 regenerated files is in the supplementary script output.

8 Conclusion

The Z-plane covariance operator of the Hilbert curve, under the spatial (face-adjacency) construction, is exactly tridiagonal Toeplitz for every dimension $D \geq 3$ and order n, with closed-form spectrum $\lambda_k = c_0 + 2c_1 \cos(\pi k/(n+1))$ (Theorem 1, Corollary 1). For the prime indicator, this spectrum correlates with the first 64 non-trivial zeta zeros at $|r|$ up to 0.990, a correlation we confirm is not an artifact of the permutation test or curve-implementation irregularities present in the original analysis pipeline (Section 7). The correlation is, however, provably incapable of becoming an equality at any order or dimension (Proposition 2): the operator's spectrum is bounded and shrinks in width as $n \to \infty$, while the zeta zero counting function grows without bound. A successful Hilbert–Pólya construction along these lines would need eigenvalue density growing as $N(T) \sim (T/2\pi)\log(T/2\pi e)$, GUE pair-correlation statistics (untested here and, based on the strongly clustered rather than repelling character of a cosine spectrum, unlikely to hold for this family), and a proof of self-adjointness in the infinite-order limit. None of these properties can be supplied by further optimizing D, n, or the choice of adjacency rule within the present construction.

Data and Code Availability

Covariance matrices (`cov_*.json`), the corrected segmented-sieve implementation, the eigenvalue re-derivation script, and the corrected permutation test are available in the companion repository. All numerical results in this paper were produced by a single script operating on the checked-in covariance matrices, run with a fixed seed, so that Tables 1–6 are regenerable without recourse to the original (Go) analysis pipeline.

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